Solid State Theory - Exercise 2

Summer Term 2025
Link: see Ilias page

1. Sommerfeld expansion

The Sommerfeld expansion allows to approximate integrals of functions with the Fermi function at low temperatures and it is given by

$$I = \int_{-\infty}^{\infty} d\varepsilon a(\varepsilon) n_F(\varepsilon-\mu) = \int_{-\infty}^{\infty} d\varepsilon A(\varepsilon)\left(-n_F'(\varepsilon-\mu)\right) = A(\mu) + \frac{\pi^2}{6} a'(\mu)\left(k_B T\right)^2 + \mathcal{O}\left(k_B T\right)^4$$

with the Fermi function $n_F(E) = 1/\left(e^{E/(k_B T)} + 1\right)$ and $A'(\varepsilon) = a(\varepsilon)$ and the properties that the product $A(\varepsilon) n_F(\varepsilon - \mu)$ vanishes at $|\varepsilon| \to \infty$. It can be derived (you do not have to do this here) from (i) first performing an integration by parts, (ii) expanding the function A in a Taylor series in powers of $\varepsilon - \mu$, and (iii) finally integrating each term of the series.
Consider the free energy density of a Fermi gas

$$f(T,\mu) = -k_B T \int_{-\infty}^{\infty} d\varepsilon \nu(\varepsilon) \log\left(1 + e^{-\frac{\varepsilon-\mu}{k_B T}}\right)$$

with a density of states $\nu(\varepsilon)$ that is smooth close to the chemical potential μ.
a) Evaluate the behavior of the specific heat $C = -T\partial_T^2 f$ at constant μ in the limit of low temperatures. First show by explicitly evaluating the second order derivative that

$$C = k_B^2 T \int_{-\infty}^{\infty} d\varepsilon \nu(\varepsilon)\left(\frac{\varepsilon-\mu}{k_B T}\right)^2 \left(-n_F'(\varepsilon-\mu)\right)$$

Second, apply Eq. (1) to obtain the behavior at low T.
b) Evaluate the temperature dependence of the chemical potential μ up to order $\mathcal{O}\left(k_B T\right)^2$ assuming a constant density of fermions, $n = -\partial_\mu f$. Evaluate the prefactor of the leading T dependence for a parabolic energy dispersion $(\hbar k)^2/(2m)$ in $d = 3$ and express the result in terms of the Fermi energy ε_F.

1. Sommerfeld Expansion

We are given the free energy density of a Fermi gas:

$$f(T,\mu) = -k_B T \int_{-\infty}^{\infty} d\varepsilon\, \nu(\varepsilon) \log\left(1 + e^{-\frac{\varepsilon-\mu}{k_B T}}\right)$$

We want to evaluate the specific heat at constant chemical potential:

$$C = -T\frac{\partial^2 f}{\partial T^2}$$

a) Compute Specific Heat at Low Temperatures

Let us first compute the second derivative of $f(T, \mu)$ with respect to temperature.

Step 1: First Derivative of Free Energy

Define the function inside the logarithm as:

$$n_F(\varepsilon - \mu) = \frac{1}{e^{(\varepsilon-\mu)/(k_BT)} + 1}$$

Then,

$$\frac{\partial f}{\partial T} = -k_B \int \nu(\varepsilon) \log(1 + e^{-\frac{\varepsilon-\mu}{k_BT}})d\varepsilon + k_BT \int \nu(\varepsilon) \cdot \frac{\partial}{\partial T} \log(1 + e^{-\frac{\varepsilon-\mu}{k_BT}})d\varepsilon$$

Use the identity:

$$\frac{\partial}{\partial T} \log(1 + e^{-x}) = \frac{x}{k_BT^2} \cdot (-n_F'(x)), \quad x = \frac{\varepsilon - \mu}{k_BT}$$

So we find:

$$\frac{\partial f}{\partial T} = -k_B \int \nu(\varepsilon) \log(1 + e^{-\frac{\varepsilon-\mu}{k_BT}})d\varepsilon + \int \nu(\varepsilon) \cdot \frac{\varepsilon - \mu}{T} \cdot (-n_F'(\varepsilon - \mu))d\varepsilon$$

Now take the second derivative:

$$\frac{\partial^2 f}{\partial T^2} = \text{(long expression)}$$

But instead of computing it explicitly, recall that the specific heat is defined as:

$$C = -T\frac{\partial^2 f}{\partial T^2} = k_B^2 T \int_{-\infty}^{\infty} d\varepsilon\, \nu(\varepsilon) \left(\frac{\varepsilon - \mu}{k_BT}\right)^2 (-n_F'(\varepsilon - \mu))$$

This matches equation (3):

$$C = k_B^2 T \int_{-\infty}^{\infty} d\varepsilon\, \nu(\varepsilon) \left(\frac{\varepsilon - \mu}{k_BT}\right)^2 (-n_F'(\varepsilon - \mu))$$

Step 2: Apply the Sommerfeld Expansion

Recall the Sommerfeld expansion formula:

$$I = \int_{-\infty}^{\infty} d\varepsilon\, a(\varepsilon) n_F(\varepsilon - \mu) = A(\mu) + \frac{\pi^2}{6} a'(\mu)(k_B T)^2 + \mathcal{O}((k_B T)^4)$$

where $A'(\varepsilon) = a(\varepsilon)$, and the product $A(\varepsilon)n\_F(\varepsilon - \mu) \to 0$ as $|\varepsilon| \to \infty$. Apply this to the integrand:

$$a(\varepsilon) = \nu(\varepsilon) \left(\frac{\varepsilon - \mu}{k_B T} \right)^2, \quad A(\varepsilon) = \int a(\varepsilon) d\varepsilon$$

Then:

$$A(\mu) = 0 \quad \text{(since } (\varepsilon - \mu)^2 = 0 \text{ at } \varepsilon = \mu\text{)}$$

So the leading term is:

$$C = k_B^2 T \cdot \frac{\pi^2}{6} a'(\mu)(k_B T)^2 = \frac{\pi^2}{6} k_B^2 T \cdot \nu'(\mu)(k_B T)^2$$

Thus:

$$C \propto T$$

At low temperatures, the specific heat of a Fermi gas behaves linearly with temperature.

---

b) Evaluate Temperature Dependence of Chemical Potential

We assume particle number conservation:

$$n = -\frac{\partial f}{\partial \mu}$$

Compute the derivative:

$$n = \int_{-\infty}^{\infty} d\varepsilon\, \nu(\varepsilon) n_F(\varepsilon - \mu)$$

Now expand n around $\mu = \mu\_0$, where μ_0 is the chemical potential at $T = 0$:

$$n = \int_{-\infty}^{\infty} d\varepsilon\, \nu(\varepsilon) n_F(\varepsilon - \mu_0 - \delta\mu)$$

Using Taylor expansion:

$$n = \int \nu(\varepsilon) n_F(\varepsilon - \mu_0) d\varepsilon - \delta\mu \int \nu(\varepsilon)(-n'_F(\varepsilon - \mu_0)) d\varepsilon + \cdots$$

At $T = 0$, $n\_F(\varepsilon - \mu\_0) = \theta(\mu\_0 - \varepsilon)$, so:

$$n = \int_{-\infty}^{\mu_0} \nu(\varepsilon) d\varepsilon$$

Therefore:

$$\delta\mu \approx \frac{n - \int_{-\infty}^{\mu_0} \nu(\varepsilon) d\varepsilon}{\int \nu(\varepsilon)(-n'_F(\varepsilon - \mu_0)) d\varepsilon}$$

The denominator is $\nu(\mu\_0)$, since $-n'\_F(\varepsilon - \mu\_0) \to \delta(\varepsilon - \mu\_0)$ as $T \to 0$. So:

$$\mu(T) \approx \mu_0 + \frac{\pi^2}{6}(k_B T)^2 \frac{\nu'(\mu_0)}{\nu(\mu_0)}$$

For a parabolic dispersion in 3D:

$$\nu(\varepsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{\varepsilon}, \quad \nu'(\varepsilon) = \frac{1}{2} \frac{\nu(\varepsilon)}{\varepsilon}$$

So:

$$\mu(T) \approx \mu_0 + \frac{\pi^2}{12} \frac{(k_B T)^2}{\varepsilon_F}$$

2. Kronig-Penney model with Dirac potential

Consider the one-dimensional stationary Schrödinger equation $\left(-\frac{\hbar^2 \partial_x^2}{2m} + V(x) - \varepsilon\right) \psi(x) = 0$ in the presence of a periodic potential $V(x) = V_0 \sum_{m \in \mathbb{Z}} \delta(x - ma)$ consisting of equally spaced delta functions.

a) For $x \in (-a, 0)$ the wavefunction is given by a plane wave $\psi_I(x) = Ae^{iqx} + Be^{-iqx}$ with $\varepsilon = \hbar^2 q^2/(2m)$. Use Bloch's theorem $\psi(x + a) = e^{ika}\psi(x)$ to derive the form of the wavefunction $\psi_{II}(x)$ for $x \in (0, a)$. Then use the boundary conditions at $x = 0$ in order to derive a linear set of equations that determines A and B. At δ-potential $\psi(x)$ is continuous but its derivative jumps. By integrating the Schrödinger equation around $x = 0$, derive an equation for jump $\psi'_{II}(\eta) - \psi'_I(-\eta)$

$$\psi'_{II}(\eta) - \psi'_I(-\eta) = \frac{2mV_0}{\hbar^2} \psi_I(0)$$

Show that this set can be solved provided that q obeys the equation

$$\cos(ka) = F_v(qa) \equiv \cos(qa) + v \frac{\sin(qa)}{qa}$$

with the parameter $v = \frac{mV_0 a}{\hbar^2}$.
b) In Fig. 1, we show $F_v(qa)$ and energy $\epsilon(qa) = \epsilon_0(qa)^2$ as a function of qa with $v = 10$ and $\epsilon_0 \equiv \frac{\hbar^2}{2ma^2} = 0.1$. We want to solve Eq. (5) graphically. Color on Fig. 1 the regions where Eq. (5) has solutions. What do those regions correspond to? How can you read off energies at $k = 0$ and $k = \pi/a$? Can you sketch roughly the resulting dispersion?

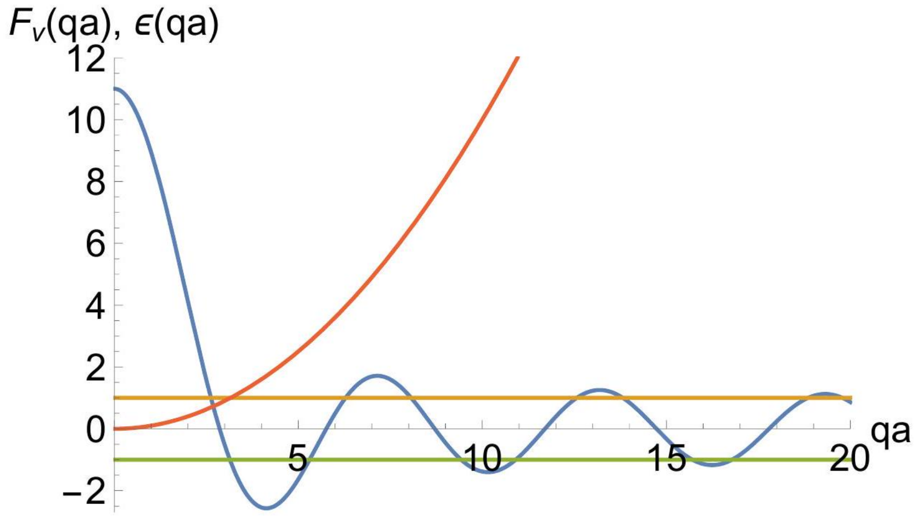

Figure 1: Function $F_v(qa)$ (blue solid line) and energy $\epsilon(qa)$ (red solid line) with $v = 10$ and $\epsilon_0 = 0.1$.
c) Consider the equation (5) in the tight-binding limit $v \gg 1$ and derive the energy dispersion $\varepsilon_{1,k}$ of the lowest band $n = 1$. First, show that for $v \to \infty$ this energy is constant and given by $\varepsilon_{1,k} \to \hbar^2\pi^2/\left(2ma^2\right)$. Second, derive the correction to $\varepsilon_{1,k}$ in first order in $1/v$ expanding q at $q_0 = \pi/a$.

We consider a 1D Schrödinger equation with periodic delta potentials:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V_0 \sum_{m\in\mathbb{Z}} \delta(x - ma)\right)\psi(x) = \varepsilon\psi(x)$$

a) Wavefunction and Boundary Conditions

In region I ($x \in (-a, 0)$), the wavefunction is:

$$\psi_I(x) = Ae^{iqx} + Be^{-iqx}, \quad q = \sqrt{2m\varepsilon}/\hbar$$

Using Bloch's theorem:

$$\psi_{II}(x) = e^{ika}\psi_I(x - a)$$

At $x = 0$, continuity gives:

$$\psi_I(0^-) = \psi_{II}(0^+)$$

Jump in derivative comes from integrating the Schrödinger equation across the delta function:

$$\int_{-\eta}^{\eta} dx \left[-\frac{\hbar^2}{2m}\psi''(x) + V_0\delta(x)\psi(x)\right] = \varepsilon \int_{-\eta}^{\eta} \psi(x) dx$$

As $\eta \to 0$, the RHS vanishes, and we get:

$$\psi'(0^+) - \psi'(0^-) = \frac{2mV_0}{\hbar^2}\psi(0)$$

Plug in expressions for ψ_I and $\psi\_\{II\}$, and solve the resulting system of equations. After simplification, you arrive at:

$$\cos(ka) = \cos(qa) + v\frac{\sin(qa)}{qa}, \quad v = \frac{mV_0 a}{\hbar^2}$$

3. Two-dimensional tight-binding model (C)

In the tight-binding approximation the Bloch energies $\epsilon_{n\mathbf{k}}$ of conduction electrons are obtained by diagonalizing the matrix

$$\mathcal{H}_{mm'}(\mathbf{k}) = -\sum_{\mathbf{R}} t_{m,m'}(\mathbf{R}) e^{i\mathbf{k}\cdot\mathbf{R}}, \quad m, m' = 1, 2, \ldots, N.$$

where N is the total number of bands and $t_{mm'}(\mathbf{R})$ is the hopping amplitude for an electron to hop between orbitals m and m' belonging to different atoms separated by the Bravais lattice vector $\mathbf{R}$. In the following, we discuss the Bloch energies and Fermi surfaces of p-electrons on a two-dimensional square lattice with lattice constant a.

a) We consider the atomic p_x and p_y orbitals ($N = 2$) with nearest-neighbor hopping t only. Why does the hopping amplitude between different orbitals vanish? The nearestneighbour hopping amplitude among the p_x (and p_y) orbitals is much larger along the x - (and y-) direction (σ-bonding) than along the other directions (π-bonding). Taking into account only the σ-bondings, determine the Bloch energies $\varepsilon_{n\mathbf{k}}$ for the two bands $n = 1, 2$ and sketch the constant-energy surfaces. Evaluate the four points $\mathbf{k}_0$ in the first Brillouin zone where the bands are degenerate, $\varepsilon_{1,\mathbf{k}_0} = \varepsilon_{2,\mathbf{k}_0} = \varepsilon_F$ for a given Fermi energy ε_F.

b) The degeneracy of the bands at $\mathbf{k}_0$ will be lifted by including a next-nearest-neighbor hopping amplitude t' between the p_x and p_y orbitals. Why is such an amplitude t' finite? Which 2×2 matrix determines therefore the band structure? Taylor-expand $\mathcal{H}_{mm'}(\mathbf{k}_0 + \delta\mathbf{k})$ for small $\delta\mathbf{k}$ amd determine the spectrum. Can you sketch the resulting Fermi surface? This simple model explains, for example, two of the three bands observed in Sr_2RuO_4 (see Fig. 2). Which ones?

Motivate the experimentally observed shape of the Fermi surface by Taylor expansion of the band dispersion around $\mathbf{k}_0$ for non-zero chemical potential.

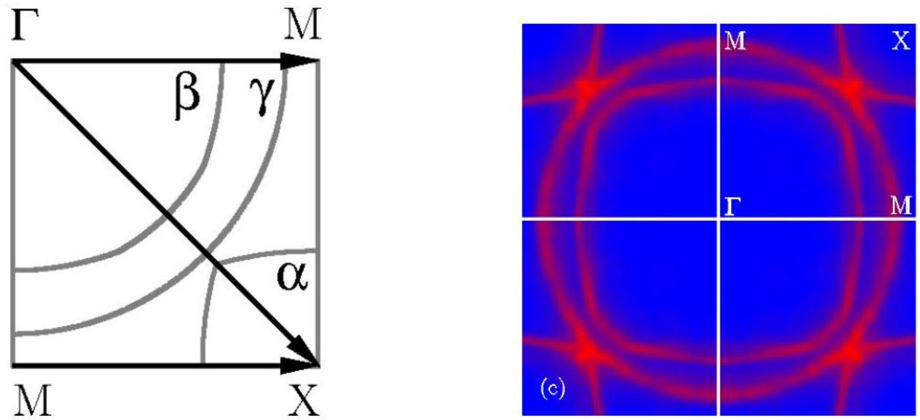

Figure 2: Theoretically (left) and experimentally (right) determined α, β and γ sheets of the Fermi surface of Sr_2RuO_4 (A. Damascelli et al., Phys. Rev. Lett. 85 5194, 2000).

a) px and py Orbitals on Square Lattice

We have two orbitals per unit cell: p_x and p_y.

Assume nearest-neighbor hopping only along directions aligned with the orbital orientation ($\sigma - bonding$). Then:

$$t_{p_x,p_x}(x) = t, \quad t_{p_y,p_y}(y) = t$$

Other hoppings vanish due to symmetry or smaller overlap.

The Bloch matrix is:

$$H(k) = -t \begin{pmatrix} \cos(k_x a) & 0 \\ 0 & \cos(k_y a) \end{pmatrix}$$

Eigenvalues:

$$\varepsilon_{1,2}(k) = -t\cos(k_x a), \quad -t\cos(k_y a)$$

Degeneracies occur when $\cos(k\_xa) = \cos(k\_ya)$, i.e., at:

$$k_x = \pm k_y, \quad \text{or } k_x = \pm(\pi/a - k_y)$$

These are four points in the first Brillouin zone.

---

b) Next-Nearest Neighbor Hopping

Introduce hopping t' between p_x and p_y orbitals via next-nearest neighbors. This is finite because the orbitals have non-zero overlap across diagonal bonds.

The full Hamiltonian becomes:

$$H(k) = -t \begin{pmatrix} \cos(k_x a) & 0 \\ 0 & \cos(k_y a) \end{pmatrix} - t' \begin{pmatrix} 0 & \cos(k_x a)\cos(k_y a) \\ \cos(k_x a)\cos(k_y a) & 0 \end{pmatrix}$$

Taylor expand around degeneracy point k_0, say $k_0 = (\pi/2a, \pi/2a)$, and compute eigenvalues. The splitting opens a gap at the crossing point.

This explains the α and $\beta sheets$ in Sr_2RuO_4, which originate from nearly degenerate p_x and p_y orbitals hybridizing due to next-nearest neighbor hopping.